

Dimension reduction for binary classification problems and tail events

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Outline

This work explores binary classification challenges in high-dimensional data. It introduces a local logistic-based dimension reduction framework as a crucial solution to mitigate the adverse effects of high-dimensional feature space.

Introduction

In standard classification problems, when dealing with high-dimensional data, it is essential to reduce the dimensionality of the original predictor space prior to conducting statistical analysis. A large number of features can significantly effect statistical analysis tools, leading to challenges in model interpretability, computational efficiency, and, in some cases, a decline in predictive performance.

Researchers have increasingly turned to dimensionality reduction to mitigate the adverse effects of high dimensions. The most relevant information is preserved by transforming the predictor space to a low-dimensional representation while redundant or noise-inducing information is discarded. As a result of this effort, the dataset is being streamlined in order to make it more amenable to statistical analysis.

Background

Consider the standard classification framework. Let $Y \in \{0, 1\}$ and $X \in \mathbb{R}^p$ and let P denote the probability measure of Y, X . We want to reduce the dimension of X while still keeping the same information to predict Y . The framework that allows for that is called sufficient dimension reduction (SDR) (Cook, 1996) and follows from the condition that

$$Y \perp X | P_\beta X$$

where $P_\beta = \beta\beta^T$ is a projection on $\text{span}(\beta)$, a (possibly) small dimension subspace of $\mathbb{R}^p$. Equivalently, it means that

$$\mathbb{P}(Y = 1|X) = \mathbb{P}(Y = 1|\beta^T X)$$

Set $d = \dim(P_\beta)$. When $d = p$, then no dimension reduction is expected and the previous condition does not imply any restriction. Whenever d is small, then estimating P_β , or equivalently the space $\text{span}(P_\beta)$ or simply a basis β of $\text{span}(P_\beta)$, becomes of prime interest in the perspective of solving the dimension curse.

How model works

Define p and p_0 such that, for almost every x in the covariate's support,

$$p(x) = \mathbb{P}(Y = 1|X = x) = p_0(\beta^T x) \quad (1)$$

Define $\text{logit}(p) = \log(p/(1-p))$ and $\text{expit}(t) = \exp(t)/(1+\exp(t))$. An important quantity in classical binary regression is $\ell(x) = \text{logit}(p(x))$. Note that

$$\nabla_x p(x) = \beta \nabla_x p_0(\beta^T x)$$

on the one hand, and

$$\nabla_x \ell(x) = -\nabla_x p(x) \frac{1}{p(x)(1-p(x))}$$

on the other hand.

Proposition: Suppose that $x \mapsto p(x)$ is differentiable. Then, for almost every x , $\nabla_x \ell(x) \in \text{span}(\beta)$.

As a consequence, being able to estimate $\nabla_x \ell(x)$ is then of interest in the aim of estimating vectors in the space $\text{span}(\beta)$.

Standard logistic

The cross-entropy loss between two discrete measure $(p, 1-p)$ and $(q, 1-q)$ is defined as

$$\text{CE}(p|q) := p \log \left(\frac{p}{q} \right) + (1-p) \log \left(\frac{1-p}{1-q} \right).$$

In our framework, we are interested in a distribution p that depends on x . So, the natural loss function is the integrated cross-entropy

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After simplification we get

$$\text{ICE}(p|q) = \mathbb{E} \left[Y \log \left(\frac{q(X)}{p(X)} \right) + (1-Y) \log \left(\frac{1-q(X)}{1-p(X)} \right) \right].$$

When the logistic link function is chosen for the parametric family, one recovers the well-known logistic regression approach.

Local logistic (proposal)

Our approach relies on “**localization**” in the previous cross-entropy loss in the covariate scale. This is crucial to estimate $\ell(x)$ and its gradient $\nabla_x \ell(x)$, after localizing around x (Fan and Gijbels, 1996).

Define the following localized version of the cross-entropy loss

$$\text{CE}_x(p|q) = \mathbb{E} \left[Y \log \left(\frac{q(X)}{p(X)} \right) + (1-Y) \log \left(\frac{1-q(X)}{1-p(X)} \right) \mid X \in B(x, \epsilon) \right].$$

The above quantity would be minimized with respect to a scalar value q , which one would naturally obtain when ϵ small, $q \simeq p(x)$

By the change of variable $q_\alpha = \text{expit}(\alpha)$ and minimizing with respect to α , one would obtain that, for ϵ small, $\alpha \simeq \text{logit}(p(x)) = \ell(x)$. Following Fan and Gijbels (1996), we have

$$q_{\alpha, \beta}(X) = \text{expit}(\alpha + \beta^T(X - x)).$$

We therefore consider the following **risk minimization** procedure

$$\arg \max_{\alpha, \beta} \{ L(\alpha, \beta, x) := \mathbb{E}[Y \log(q_{\alpha, \beta}(X)) + (1-Y) \log(1 - q_{\alpha, \beta}(X)) \mid X \in B(x, \epsilon)] \} \quad (2)$$

For estimation of coefficients, further practice the localization-based **empirical risk minimization** procedure

$$(\hat{\alpha}(x), \hat{\beta}(x)) = \arg \max_{\alpha, \beta} L_n(\alpha, \beta, x) + \lambda \|\beta\|_1 \quad (3)$$

with

$$L_n(\alpha, \beta, x) = \sum_{i \in N_k(x)} Y_i \log(q_{\alpha, \beta}(z_i)) + (1-Y_i) \log(1 - q_{\alpha, \beta}(z_i))$$

Aggregating the directions

The expression (3) allows to estimate $\nabla_x \ell(x)$, and this should be done for several x in order to obtain many vectors in the dimension reduction space of interest. Define

$$M = \int_{\mathbb{R}^p} \nabla_x \ell(x) \nabla_x \ell(x)^T f_X(x) dx$$

which is estimated by

$$\hat{M} = (1/m) \sum_{i=1}^m \hat{\beta}(X_i) \hat{\beta}(X_i)^T$$

where X_i are chosen uniformly random.

Algorithm

In order to estimate M , we use algorithm 1.

Algorithm 1 Algorithm developed for estimation of M .

- 1: **Require:** $y \in \{0, 1\}$, $\mathbf{X} \in \mathbb{R}^p$.
- 2: Select m samples without replacement from $\mathbf{X}$.
- 3: **for** each $x \in m$ **do**
- 4: $N_k(x) \leftarrow$ Search $k = \sqrt{n}$ nearest neighbours from X .
- 5: Select instances of y : $y_{N_k(x)} \leftarrow N_k(x)$
- 6: Centralize searched nearest neighbors: $N_k(x) = z - x_m$.
- 7: Estimate α using β using (3).
- 8: Store in M matrix for each $x \in m$.
- 9: **end for**.
- 10: Return $\hat{M}$.

Synthetic data example

We simulated a vector of covariates $\mathbf{X} \in \mathbb{R}^p$, $p = 100$ of size $n = 1000$ from $\mathcal{N}(0, 1)$.

The response variable y is a Bernoulli random variable and simulated by using the following parameter $p = p_1 p_2 \in (0, 1)$. Where $p_1 = \text{expit}(X^T \beta_1)$ with $\beta_1 = (1, 1, \text{rep}(0, 99))$ and $p_2 = \text{expit}(X^T \beta_2)$ with $\beta_2 = (1, 0, 1, \text{rep}(0, 98))$.

Results

We apply our developed **knn** based cross-validation forward selection procedure.

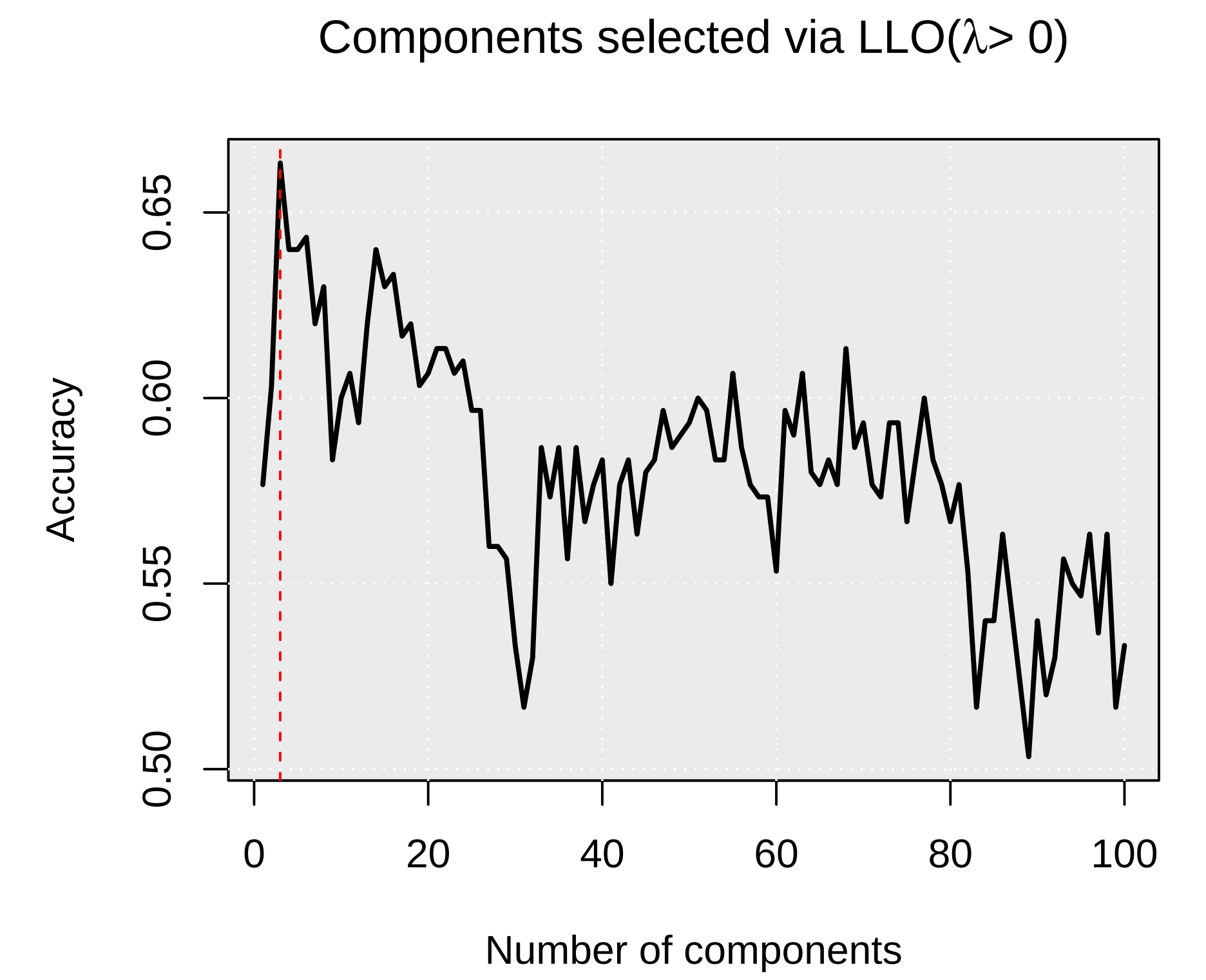


Figure 1. Important features selection through cross-validation for LLO($\lambda > 0$)

We split the data into training and testing pockets and applied **knn** classifier. Table 1 provides the assessment measures results.

Table 1. AM risk, misclassification risk, the area under the curve, and computing time of the models when using **knn** classifier.

Model	AM risk	Miscl risk	AUC	Est time
Full data model	0.487	0.487	0.497	2.56
Oracle model	0.337	0.337	0.734	0.68
LLO ($\lambda = 0$)	0.293	0.293	0.761	0.75
LLO ($\lambda > 0$)	0.287	0.287	0.746	0.69
LLO ($\lambda = 0$, <i>fps</i>)	0.517	0.517	0.501	2.68
LLO ($\lambda > 0$, <i>fps</i>)	0.493	0.493	0.509	2.43

The ROC curves in Figure 2 reveal that the models employed with a reduced predictor space are very similar to the Oracle model and represent the top-performing models in a given classification problem.

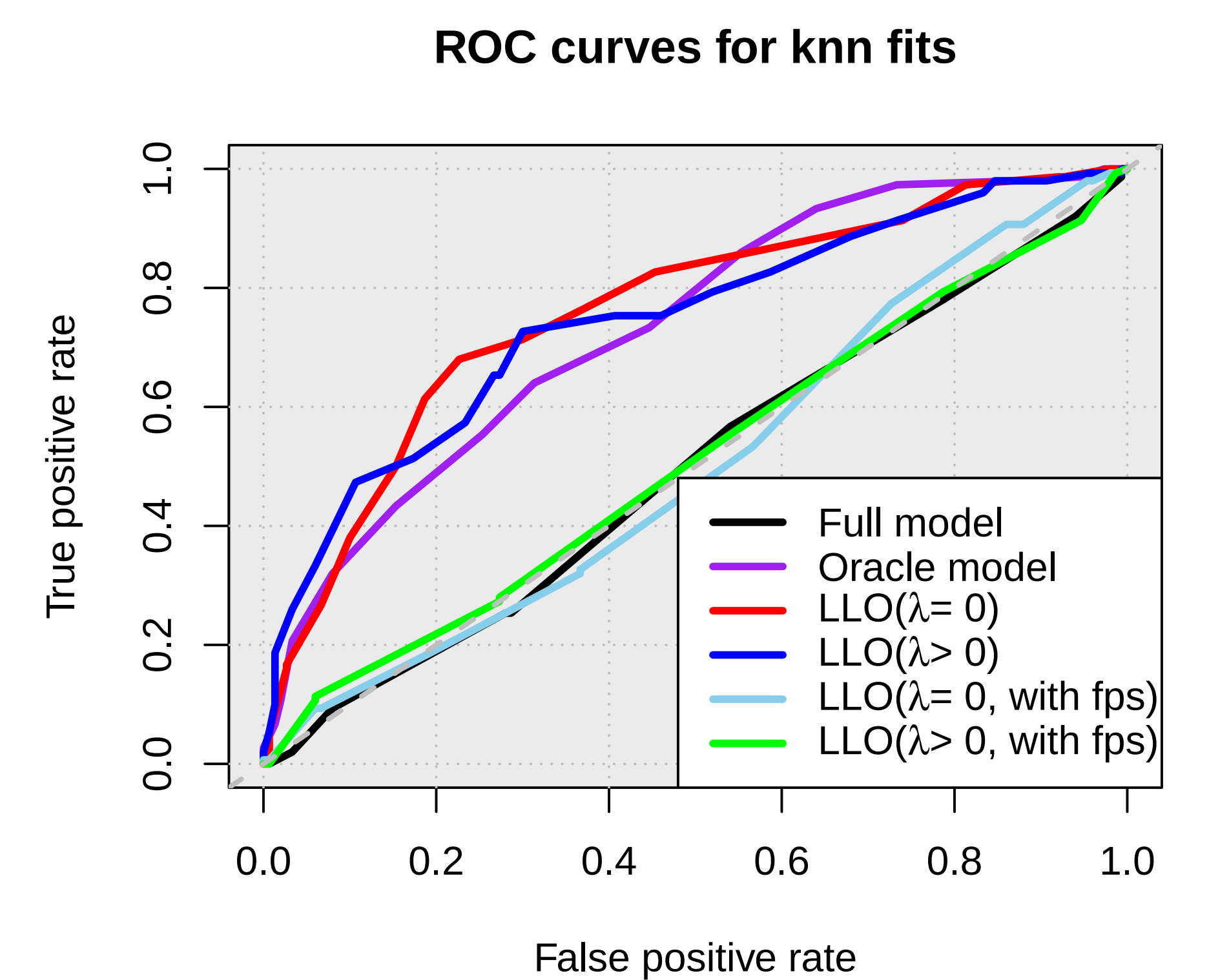


Figure 2. ROC curves of the fitted models

Ongoing work

- Dimension reduction for the tail event.

$$p(t|x) = \mathbb{P}(Y > t|X = x)$$

where t is a sufficiently large threshold and defines the tail events

References

Cook, R. D. (1996). Graphics for Regressions With a Binary Response. *Journal of the American Statistical Association*, 91(435), 983–992. <https://doi.org/10.2307/2291717>

Fan, J. and I. Gijbels (1996). *Local polynomial modelling and its applications*, Volume 66 of *Monographs on Statistics and Applied Probability*. Chapman & Hall, London